

STAT525 HOMEWORK#5

1. Use the **patient satisfaction** data described in KNNL Problem 6.15.
 - a Compute the pairwise correlations between the X 's and between each X and Y . Which X variable appears to be the best individual predictor?
 - b Run the linear regression with age, severity of illness and anxiety level as the explanatory variables and satisfaction as the response variable. Summarize the regression results.
 - c Plot the residuals versus the predicted satisfaction and each of the explanatory variables. Are there any unusual patterns?
 - d Examine the assumption of normality for the residuals using a qqplot or histogram. State your conclusions.
 - e Predict the satisfaction for a 55 year old patient with illness severity 50 and anxiety level 2.8. Provide a 95% prediction interval with your prediction.
2. KNNL Problem 7.5
3. KNNL Problem 7.6 (Please try to only use SAS statement `model y = x1 x2 x3/ss1;`, and derive the test based on it. Then justify your answer with `test x2 x3;`)
4. Derive the equation (7.56) on page 281 (HINT: Recall that b_1 can be obtained by regressing the residuals of $Y|X_2$ vs the residuals of $X_1|X_2$) (STAT Majors)
5. KNNL Problem 8.6
6. KNNL Problem 8.15

cli - Confidence Limits for Individual (Prediction Interval)
 clm - Confidence Limits for Mean (CI for Mean Response)
 clb - Confidence Limits for Regression Coefficients

1. Use the **patient satisfaction** data described in KNNL Problem 6.15.

- a Compute the pairwise correlations between the X 's and between each X and Y .
Which X variable appears to be the best individual predictor?

```
61 /*****  
62 * Compute pairwise correlations between X's and between each X and Y  
63 *****/  
64  
65 proc corr data=a1 noprob;  
66     var x1 x2 x3 y;  
67 run;
```

The CORR Procedure

4 Variables: x1 x2 x3 y

Simple Statistics						
Variable	N	Mean	Std Dev	Sum	Minimum	Maximum
x1	46	61.56522	17.23646	2832	26.00000	92.00000
x2	46	38.39130	8.91809	1766	22.00000	55.00000
x3	46	50.43478	4.31356	2320	41.00000	62.00000
y	46	2.28696	0.29934	105.20000	1.80000	2.90000

Pearson Correlation Coefficients, N = 46				
	x1	x2	x3	y
x1	1.00000	-0.78676	-0.60294	-0.64459
x2	-0.78676	1.00000	0.56795	0.56968
x3	-0.60294	0.56795	1.00000	0.67053
y	-0.64459	0.56968	0.67053	1.00000

• The strongest correlation with Y is X_3 (Anxiety Level), $r=0.671$

• This suggests X_3 (Anxiety Level) is the best predictor for patient satisfaction.

• However, correlations among the X 's show moderate collinearity.
Ex: X_2 vs X_1 , $r=-0.79$

- b Run the linear regression with age, severity of illness and anxiety level as the explanatory variables and satisfaction as the response variable. Summarize the regression results.

```
66 /*****  
67 * Multiple Regression: Y = X1 + X2 + X3  
68 *****/  
69  
70 proc reg data=a1;  
71     model y = x1 x2 x3;  
72 run;
```

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read 46
Number of Observations Used 46

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	2.18362	0.72787	16.54	<.0001
Error	42	1.84856	0.04401		
Corrected Total	45	4.03217			

Root MSE 0.20979 R-Square 0.5415
Dependent Mean 2.28696 Adj R-Sq 0.5088
Coeff Var 9.17348

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1.05325	0.61379	1.72	0.0935
x1	1	-0.00586	0.00309	-1.90	0.0647
x2	1	0.00193	0.00579	0.33	0.7407
x3	1	0.03015	0.00926	3.26	0.0022

$$\hat{Y} = 1.053 - 0.0059X_1 + 0.0019X_2 + 0.0302X_3$$

• The overall model is significant ($F=16.54$, $p<0.0001$), indicating that age, severity, and anxiety jointly explain the variation in patient satisfaction.

• $R^2=0.54$: 54% variation in patient satisfaction is explained by age, severity, and anxiety.

• Age (X_1) has a negative but marginally significant effect ($p=0.065$), indicating that older patients tend to report slightly lower patient satisfaction.

• Severity of Illness (X_2) not significant ($p=0.74$).

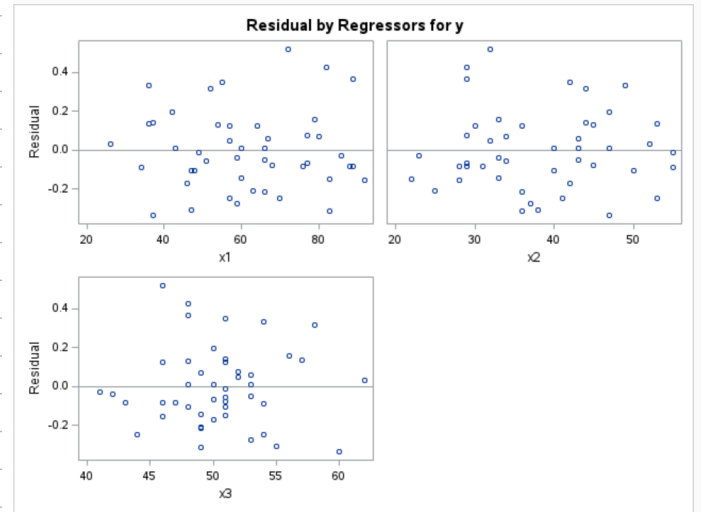
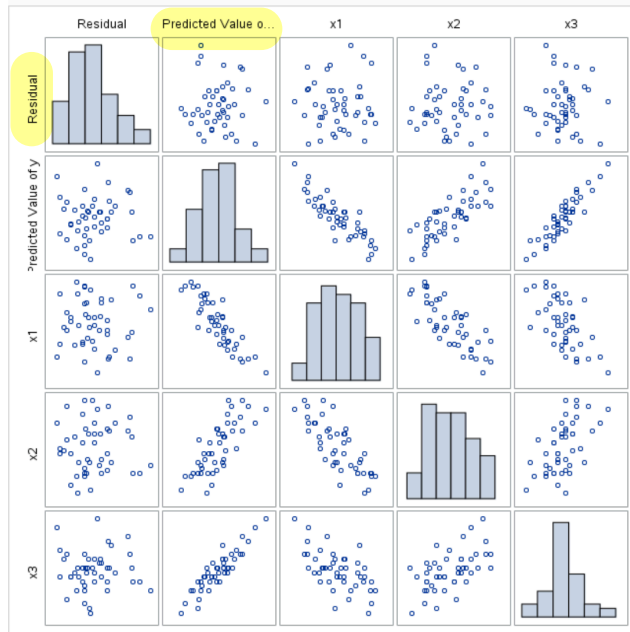
• Anxiety Level (X_3) is significant, indicating that higher anxiety tend to report higher patient satisfaction.

c Plot the residuals versus the predicted satisfaction and each of the explanatory variables. Are there any unusual patterns?

```

76 /*****
77 * Residuals versus predicted and explanatory variables
78 *****/
79 proc sgscatter data=out1;
80     matrix resid pred x1 x2 x3 / diagonal=(histogram);
81 run;

```



• Residual Plots seems to all be scattering around zero across $\hat{y}$ and x_1, x_2, x_3 .

• No obvious curvature or funnel pattern, so linearity and constant variance may be satisfied.

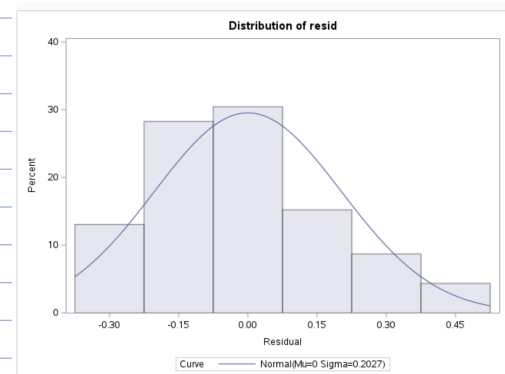
• No major outliers observed.

d Examine the assumption of normality for the residuals using a qqplot or histogram. State your conclusions.

```

89 /*****
90 * Normality Check for Residuals
91 *****/
92 proc univariate data=out1 normal;
93     var resid;
94     histogram resid / normal;
95     qqplot resid / normal(mu=est sigma=est);
96 run;

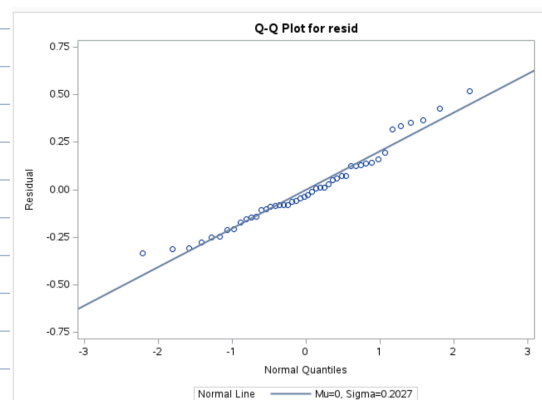
```



Residuals appear to follow normal distribution, no serious violations of the model's normality assumption.

• Histogram of e_i approx. bell-shaped and centered near 0, matching fitted normal curve well.

• Q-Q Plot points lie approx. along the diagonal referenced line with only minor deviations at tails.



e Predict the satisfaction for a 55 year old patient with illness severity 50 and anxiety level 2.8. Provide a 95% prediction interval with your prediction.

```

93 /*****
94 * Prediction and 95% Prediction Interval
95 *****/
96 /* Step 1: New observation */
97 data new;
98     input x1 x2 x3;
99     datalines;
100 55 50 2.8
101 ;
102 run;
103
104 /* Step 2: Combine original data + new observation */
105 data combined;
106     set al new;
107 run;
108
109 /* Step 3: Fit model and compute prediction + 95% PI */
110 proc reg data=combined;
111     model y = x1 x2 x3 / cli;
112     id x1 x2 x3;
113     output out=pred p=predicted lcl=lcl95 ucl=ucl95;
114 run;
115 quit;
116
117 /* Step 4: Print only the new case */
118 proc print data=pred;
119     where missing(y); /* this filters to the new observation (since y is missing) */
120     var x1 x2 x3 predicted lcl95 ucl95;
121     title "Predicted Satisfaction and 95% Prediction Interval for New Patient";
122 run;

```

Predicted Satisfaction and 95% Prediction Interval for New Patient

Obs	x1	x2	x3	predicted	lcl95	ucl95
47	55	50	2.8	0.91173	-0.11576	1.93923

• Patient Satisfaction level when 55 year old with illness severity 50 and anxiety 2.8 = 0.912, with PI: (-0.11576, 1.93923)

2. KNNL Problem 7.5

*7.5. Refer to Patient satisfaction Problem 6.15.

a. Obtain the analysis of variance table that decomposes the regression sum of squares into extra sums of squares associated with X_2 ; with X_1 , given X_2 ; and with X_3 , given X_2 and X_1 .

(Sequential Type I)
order: X_2, X_1, X_3

```

125 /*****
126 * 2a: Obtain extra SS for X2, X1|X2, and X3|X2,X1
127 *****/
128 /* Order 1: X2 entered first, then X1, then X3 */
129 title "Decomposition of Regression SS into Extra Sums of Squares";
130 proc reg data=al;
131     model y = x2 x1 x3 / ssl ss2;
132 run;

```

Decomposition of Regression SS into Extra Sums of Squares

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	46
Number of Observations Used	46

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	2.18362	0.72787	16.54	<.0001
Error	42	1.84856	0.04401		
Corrected Total	45	4.03217			

Root MSE	0.20979	R-Square	0.5415
Dependent Mean	2.28696	Adj R-Sq	0.5088
Coeff Var	9.17348		

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Type I SS	Type II SS
Intercept	1	1.05325	0.61379	1.72	0.0935	240.58783	0.12960
x2	1	0.00193	0.00579	0.33	0.7407	1.30857	0.00489
x1	1	-0.00586	0.00309	-1.90	0.0647	0.40818	0.15844
x3	1	0.03015	0.00926	3.26	0.0022	0.46686	0.46686

$$SS = SS(X_2) + SS(X_1|X_2) + SS(X_3|X_1, X_2)$$

$$• SS(X_2) = 1.30857$$

$$• SS(X_1|X_2) = 0.40818$$

$$• SS(X_3|X_1, X_2) = 0.46686$$

Add up to: 2.18362

- b. Test whether X_3 can be dropped from the regression model given that X_1 and X_2 are retained. Use the F^* test statistic and level of significance $.025$. State the alternatives, decision rule, and conclusion. What is the P -value of the test?

$H_0: \beta_3 = 0$ (X_3 doesn't contribute additional explanatory power)
 $H_A: \beta_3 \neq 0$ (X_3 contribute additional explanatory power after accounting for $X_1 + X_2$)

GLM in terms of Extra SS:

$$F^* = \frac{SS(X_3|X_1, X_2)/(3-2)}{MSE}$$

Reject H_0 . Therefore, X_3 contribute additional explanatory power, even after accounting for $X_1 + X_2$.

$$F^* = \frac{0.46686/1}{0.09901} = 10.61 \sim F_{1,42}^* > 5.17 \sim F_{0.025, 1, 42}$$

$$p\text{-value} = 0.0022 < 0.025$$

one-way

3. KNNL Problem 7.6 (Please try to only use SAS statement model $y = x_1 \ x_2 \ x_3 / ss1;$, and derive the test based on it. Then justify your answer with test $x_2 \ x_3;$)

- *7.6. Refer to Patient satisfaction Problem 6.15. Test whether both X_2 and X_3 can be dropped from the regression model given that X_1 is retained. Use $\alpha = .025$. State the alternatives, decision rule, and conclusion. What is the P -value of the test?

```
134 /*****
135 * 3: Test contribution of X2 and X3
136 *****/
137 proc reg data=a1;
138     model y = x1 x2 x3 / ss1;
139     test x2, x3;
140 run;
```

The REG Procedure
 Model: MODEL1
 Dependent Variable: y

Number of Observations Read	46
Number of Observations Used	46

Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Value
Model	3	2.18362	0.72787	16.54
Error	42	1.84856	0.04401	
Corrected Total	45	4.03217		

Root MSE	0.20979	R-Square	0.5415
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Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1.05325	0.61379	1.72	0.0935
x1	1	-0.00586	0.00309	-1.90	0.0647
x2	1	0.00193	0.00579	0.33	0.7407
x3	1	0.03015	0.00926	3.26	0.0022

$H_0: \beta_2 = \beta_3 = 0$
 $H_A: \text{At least one } \beta_2 \text{ or } \beta_3 \neq 0$

GLM in terms of Extra SS:

$$F^* = \frac{SS(X_1, X_2 | X_3)/(3-1)}{MSE}$$

$$F^* = \frac{0.50826/2}{0.04901} = 5.17 \sim F_{2,42}^* > 3.23 \sim F_{0.025, 2, 42}$$

$$p\text{-value} = 0.0067 < 0.025$$

one-way

Reject H_0 . Therefore, At least one X_2 or X_3 contribute additional explanatory power, even after accounting for X_1 .

$$SS(X_1, X_2 | X_3) = 0.04139 + 0.46686 = 0.50826$$

4. Derive the equation (7.56) on page 281 (HINT: Recall that b_1 can be obtained by regressing the residuals of $Y|X_2$ vs the residuals of $X_1|X_2$)

(skipped)

To show that the regression coefficient of X_1 is unchanged when X_2 is added to the regression model in the case where X_1 and X_2 are uncorrelated, consider the following algebraic expression for b_1 in the first-order multiple regression model with two predictor variables:

$$b_1 = \frac{\frac{\sum(X_{1i} - \bar{X}_1)(Y_i - \bar{Y})}{\sum(X_{1i} - \bar{X}_1)^2} - \left[\frac{\sum(Y_i - \bar{Y})^2}{\sum(X_{1i} - \bar{X}_1)^2} \right]^{1/2} r_{Y2}r_{12}}{1 - r_{12}^2} \quad (7.56)$$

Step-by-Step Derivation of Equation (7.56)

Step 1: Start with a multiple regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$$

We want to derive an expression for β_1 (denoted as b_1 in sample terms).

Step 2: Apply the FWL Theorem

According to FWL, you can estimate b_1 by:

$$b_1 = \frac{\sum(X_{1i}^* - \bar{X}_1^*)(Y_i^* - \bar{Y}^*)}{\sum(X_{1i}^* - \bar{X}_1^*)^2}$$

Where:

- X_1^* = residuals from regressing X_1 on X_2
- Y^* = residuals from regressing Y on X_2

Step 3: Use residual properties

These residuals can be standardized, and the sample covariance becomes a function of correlation coefficients.

Let:

- r_{12} : correlation between X_1 and X_2
- r_{Y2} : correlation between Y and X_2

Then:

$$b_1 = \frac{S_{X_1 Y} - r_{Y2} r_{12} S_Y}{S_{X_1} (1 - r_{12}^2)}$$

Rewriting in terms of sums of squares:

Step 4: Equation (7.56)

$$b_1 = \frac{\sum(X_{1i} - \bar{X}_1)(Y_i - \bar{Y}) - \left[\frac{\sum(Y_i - \bar{Y})^2}{\sum(X_{1i} - \bar{X}_1)^2} \right]^{1/2} \cdot r_{Y2} r_{12}}{1 - r_{12}^2}$$

This matches equation (7.56) exactly.

Interpretation:

If $r_{12} = 0$, then the second term becomes 0, and:

$$b_1 = \frac{\sum(X_{1i} - \bar{X}_1)(Y_i - \bar{Y})}{\sum(X_{1i} - \bar{X}_1)^2}$$

which is the simple linear regression coefficient for regressing Y on X_1 alone.

Hence, when X_1 and X_2 are uncorrelated, the coefficient of X_1 does not change whether or not X_2 is included.

5. KNNL Problem 8.6

8.6. Steroid level. An endocrinologist was interested in exploring the relationship between the level of a steroid (Y) and age (X) in healthy female subjects whose ages ranged from 8 to 25 years. She collected a sample of 27 healthy females in this age range. The data are given below:

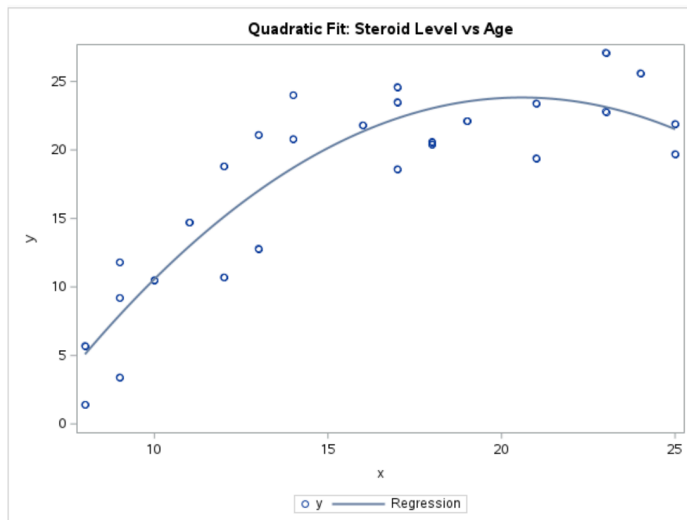
i :	1	2	3	...	25	26	27
X_i :	23	19	25	...	13	14	18
Y_i :	27.1	22.1	21.9	...	12.8	20.8	20.6

- a. Fit regression model (8.2). Plot the fitted regression function and the data. Does the quadratic regression function appear to be a good fit here? Find R^2 .

```

184 proc reg data=steroid;
185     model y = x x2 / clb;
186     plot y*x;
187     title "Quadratic Regression Model: Steroid Level vs Age";
188 run;
189
190 proc sgplot data=steroid;
191     scatter x=x y=y;
192     reg x=x y=y / degree=2;
193     title "Quadratic Fit: Steroid Level vs Age";
194 run;

```



Regression seems to be fitting the data well, showing clear curvature.

Quadratic Regression Model: Steroid Level vs Age

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	27
Number of Observations Used	27

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	1046.26586	523.13293	52.63	<.0001
Error	24	238.54081	9.93920		
Corrected Total	26	1284.80667			

Root MSE	3.15265	R-Square	0.8143
Dependent Mean	17.64444	Adj R-Sq	0.7989
Coeff Var	17.86766		

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	95% Confidence Limits
Intercept	1	-26.32541	5.88154	-4.48	0.0002	-38.46431 -14.18652
x	1	4.87357	0.77515	6.29	<.0001	3.27375 6.47340
x2	1	-0.11840	0.02347	-5.05	<.0001	-0.16684 -0.06996

$$\hat{\text{Steroid}} = -26.32541 + 4.87357 \text{age} - 0.11840 \text{age}^2$$

$R^2 = 0.8143$, 81.43% of variability in steroid level is explained by age and age².

- b. Test whether or not there is a regression relation; use $\alpha = .01$. State the alternatives, decision rule, and conclusion. What is the P -value of the test? overall

$$H_0: \beta_1 = \beta_2 = 0$$

$$H_A: \text{At least one } \beta_1 \text{ or } \beta_2 \neq 0$$

$$F = 52.63$$

$$df_1 = 2$$

$$df_2 = 24$$

$$p\text{-value} < 0.0001 \ll 0.01 = \alpha$$

$$\alpha = 0.01$$

Reject H_0 . Therefore, At least one β_1 or β_2 is a significant predictor. So there is a significant relationship between age and steroid level.

- c. Obtain joint interval estimates for the mean steroid level of females aged 10, 15, and 20, respectively. Use the most efficient simultaneous estimation procedure and a 99 percent family confidence coefficient. Interpret your intervals.

```

67 /* Step 1: Create new input values */
68 data new;
69   input x;
70   x2 = x*x;
71   y = .;
72   datalines;
73 10
74 15
75 20
76 ;
77 run;
78
79 /* Step 2: Combine original + new data */
80 data all;
81   set steroid new;
82 run;
83
84 /* Step 3: Fit model & output predicted values + standard error */
85 proc reg data=all;
86   model y = x x2 / alpha=0.01 clm; → Mean
87   id x y x2;
88   output out=ci_out p=pred stdp=stderr;
89 run;
90
91 /* Step 4: Compute Working-Hotelling 99% joint confidence intervals */
92 data joint_CI;
93   set ci_out;
94   if y = .; /* only new input values */
95   fval = finv(0.99, 2, 24); /* 2 predictors, n=27 → df = n - p - 1 = 24 */
96   WH = sqrt(2 * fval);
97   lowerWH = pred - WH * stderr;
98   upperWH = pred + WH * stderr;
99   keep x pred stderr lowerWH upperWH;
100 run;
101
102 proc print data=joint_CI noobs;
103   title "Working-Hotelling 99% Joint Confidence Intervals for E(y)";
104 run;

```

Working-Hotelling (WH) best for estimating multiple means

$$\hat{y}_h \pm W \cdot SE(\hat{y}_h)$$

- $\hat{y}_h$ = predicted mean at age x_h
- $W = \sqrt{2 \cdot F_{2, n-3, 0.01}} \rightarrow \alpha$
- $SE(\hat{y}_h) = \sqrt{MSE} \cdot \mathbf{h}_h$, where $\mathbf{h}_h = \mathbf{x}_h' (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_h$
- $\mathbf{X}$ includes intercept, x , and x^2

Using SAS output:

$$W = \sqrt{2 \cdot F_{2, 24, 0.01}} = 3.4$$

Working-Hotelling 99% Joint Confidence Intervals for E(y)

x	pred	stderr	lowerWH	upperWH
10	10.5702	0.92384	7.4747	13.6657
15	20.1379	0.89276	17.1465	23.1293
20	23.7856	0.85753	20.9122	26.6589

Intervals account for "Joint Estimation Uncertainty" over all 3 X-values, using WH.

- We are 99% confident that the true mean steroid level at:
 - Age 10 lies between 7.47~13.67
 - Age 15 lies between 17.15~23.13
 - Age 20 lies between 20.91~26.66

- d. Predict the steroid levels of females aged 15 using a 99 percent prediction interval. Interpret your interval.

```

107 data new;
108   input x;
109   x2 = x*x;
110   y = .;
111   datalines;
112 15
113 ;
114 run;
115
116 data all;
117   set steroid new;
118 run;
119
120 proc reg data=all;
121   model y = x x2 / alpha=0.01 cli; /* cli gives prediction interval */
122   id x;
123 run;

```

Individual

• 15 Year-old female, predicted steroid level 20.14

• 99% PI for a randomly selected 15 Year-old (10.97, 29.30)

The REG Procedure
Model: MODEL1
Dependent Variable: y

Output Statistics							
Obs	x	Dependent Variable	Predicted Value	Std Error Mean Predict	99% CL Predict	Residual	
1	23	27.1	23.1325	1.0721	13.8188 32.4462	3.9675	
2	19	22.1	23.5297	0.8804	14.3745 32.6848	-1.4297	
3	25	21.9	21.5132	1.6078	11.6149 31.4114	0.3868	
4	12	10.7	15.1077	0.7813	6.0232 24.1922	-4.4077	
5	8	1.4	5.0855	1.4207	-4.5862 14.7572	-3.6855	
6	12	18.8	15.1077	0.7813	6.0232 24.1922	3.6923	
7	11	14.7	12.9574	0.8106	3.8528 22.0619	1.7426	
8	8	5.7	5.0855	1.4207	-4.5862 14.7572	0.6145	
9	17	18.6	22.3074	0.9223	13.1200 31.4948	-3.7074	
10	18	20.4	23.0369	0.9072	13.8613 32.2125	-2.6369	
11	9	9.2	7.9463	1.1301	-1.4209 17.3134	1.2537	
12	21	23.4	23.8047	0.8635	14.6621 32.9473	-0.4047	
13	10	10.5	10.5702	0.9238	1.3816 19.7588	-0.0702	
14	25	19.7	21.5132	1.6078	11.6149 31.4114	-1.8132	
15	9	11.8	7.9463	1.1301	-1.4209 17.3134	3.8537	
16	17	24.6	22.3074	0.9223	13.1200 31.4948	2.2926	
17	9	3.4	7.9463	1.1301	-1.4209 17.3134	-4.5463	
18	23	22.8	23.1325	1.0721	13.8188 32.4462	-0.3325	
19	13	21.1	17.0212	0.8062	7.9197 26.1228	4.0788	
20	14	24.0	18.6980	0.8513	9.5644 27.8315	5.3020	
21	16	21.8	21.3411	0.9180	12.1571 30.5251	0.4589	
22	17	23.5	22.3074	0.9223	13.1200 31.4948	1.1926	
23	21	19.4	23.8047	0.8635	14.6621 32.9473	-4.4047	
24	24	25.6	22.4413	1.3010	12.9022 31.9803	3.1587	
25	13	12.8	17.0212	0.8062	7.9197 26.1228	-4.2212	
26	14	20.8	18.6980	0.8513	9.5644 27.8315	2.1020	
27	18	20.6	23.0369	0.9072	13.8613 32.2125	-2.4369	
28	15	.	20.1379	0.8928	10.9734 29.3024	.	

- e. Test whether the quadratic term can be dropped from the model; use $\alpha = .01$. State the alternatives, decision rule, and conclusion.

$H_0: \beta_2 = 0$ (Quadratic can be dropped)
 $H_A: \beta_2 \neq 0$ (Quadratic can't be dropped)

From SAS output:

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	95% Confidence Limits
Intercept	1	-26.32541	5.88154	-4.48	0.0002	-38.46431 -14.18652
x	1	4.87357	0.77515	6.29	<.0001	3.27375 6.47340
x2	1	-0.11840	0.02347	-5.05	<.0001	-0.16684 -0.06996

Reject H_0 . Therefore, there is strong evidence that the X_2^2 Quadratic term cannot be dropped and is significant to the model.

p-value: < 0.0001 << 0.01 = α

- f. Express the fitted regression function obtained in part (a) in terms of the original variable X .

$$\hat{y} = -26.32541 + 4.87357X - 0.11840X^2 + \epsilon$$

6. KNNL Problem 8.15

- 8.15. Refer to Copier maintenance Problem 1.20. The users of the copiers are either training institutions that use a small model, or business firms that use a large, commercial model. An analyst at Tri-City wishes to fit a regression model including both number of copiers serviced (X_1) and type of copier (X_2) as predictor variables and estimate the effect of copier model (S—small, L—large) on number of minutes spent on the service call. Records show that the models serviced in the 45 calls were:

i:	1	2	3	...	43	44	45
X_{i2} :	S	L	L	...	L	L	L

Assume that regression model (8.33) is appropriate, and let $X_2 = 1$ if small model and 0 if large, commercial model.

- a. Explain the meaning of all regression coefficients in the model.

$$\hat{y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

y : Minutes spent on service call

X_1 : Number of copiers serviced

X_2 : Copier model (1=small, 0=large)

β_0 : Expected Service Time (in minutes) for a large commercial model, ($X_2 = 0$), when 0 copiers are serviced ($X_1 = 0$)
 • Baseline

β_1 : Increase in Service Time (in minutes) for each additional copiers serviced, adjusting for copier time.

β_2 : Difference in Service Time (in minutes) between small and large models, controlling for the number of copiers serviced.

b. Fit the regression model and state the estimated regression function.

```
183 proc reg data=copier;
184     model y = x1 x2 / clb;
185     title "Regression Model: Service Time vs Number of Copiers and Model Type";
186 run;
187 quit;
```

Regression Model: Service Time vs Number of Copiers and Model Type

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	45
Number of Observations Used	45

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	77001	38501	479.04	<.0001
Error	42	3375.53856	80.36997		
Corrected Total	44	80377			

Root MSE	8.96493	R-Square	0.9580
Dependent Mean	76.26667	Adj R-Sq	0.9560
Coeff Var	11.75472		

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	95% Confidence Limits
Intercept	1	-1.37162	3.03082	-0.45	0.6532	-7.48806 4.74481
x1	1	15.05150	0.48641	30.94	<.0001	14.06989 16.03312
x2	1	1.99232	2.79494	0.71	0.4799	-3.64809 7.63273

$$\hat{y} = -1.37162 + 15.05150x_1 + 1.99232x_2$$

c. Estimate the effect of copier model on mean service time with a 95 percent confidence interval. Interpret your interval estimate.

$$95\% \text{ CI: } (-3.64809, 7.63273)$$

• We are 95% confident that β_2 : Difference in Service Time (in minutes) between small and large models, controlling for the number of copiers serviced, lies between -3.65 and 7.63 minutes.

• Since this interval contains 0, we fail to reject H_0 , therefore cannot conclude copier model predictor is significant.

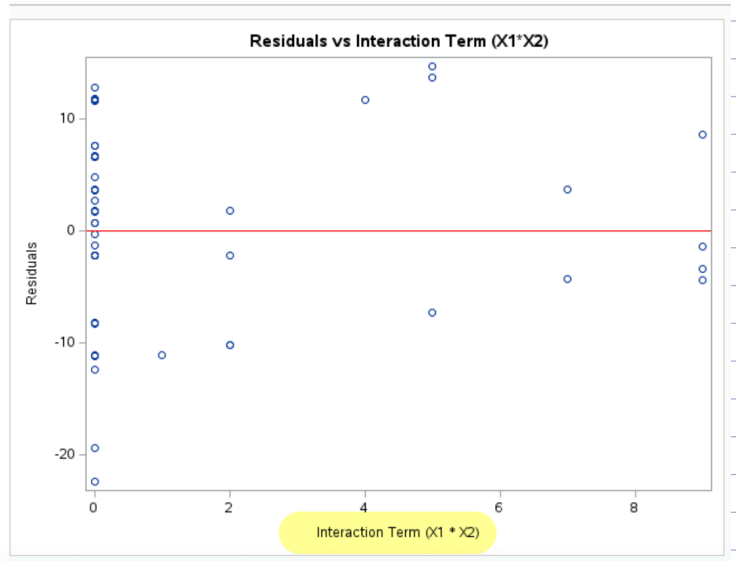
d. Why would the analyst wish to include X_1 , number of copiers, in the regression model when interest is in estimating the effect of type of copier model on service time?

• Include X_1 as a possible confounder; may also affect total service time regardless of model type.

• X_1 controls for number of copiers serviced, which directly affects service time, making sure model type would not be confounded by differences in workload.

e. Obtain the residuals and plot them against $X_1'X_2$. Is there any indication that an interaction term in the regression model would be helpful?

```
248 /* Fit model and output residuals */
249 proc reg data=copier;
250     model y = x1 x2;
251     output out=residout r=resid;
252 run;
253 quit;
254
255 /* Plot residuals vs interaction term */
256 proc sgplot data=residout;
257     scatter x=x1x2 y=resid;
258     refline 0 / axis=y lineattrs=(color=red);
259     xaxis label="Interaction Term (X1 * X2)";
260     yaxis label="Residuals";
261     title "Residuals vs Interaction Term (X1*X2)";
262 run;
```



◦ Noticable Spread and variability at certain interaction values, suggesting the relationship between X_1 and $\hat{y}$ might depend on X_2 .

◦ Adding interaction would improve fit.